

Reviewer Pack — Minimal Rank of Tensor Product of Tropicalized Matrices Boun...

Entry 019b1dc02678 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Tensor Product of Tropicalized Matrices Bounds ACC⁰ Circuit Size

Entry ID: 019b1dc02678 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-27 17:09:48 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry (Tensor Products) **Field B** (complexity object): Complexity Theory: ACC⁰ Circuit Complexity

Statement:

{‘text’: ‘For every explicit function $f: \{0, \dots, N-1\} \rightarrow \mathbb{R}$ in $\mathcal{P}$, let T_f be the tropicalization of the matrix A_f whose rows are the evaluations of f at powers of a fixed primitive element. Define the minimal rank of the tensor product of the columns of T_f as $R_{\min}(T_f)$. Conjecture: If f is computable in subexponential time, then $R_{\min}(T_f) = \Omega(\log^2 n)$, where n is the number of variables in f .’, ‘invariant’: ‘ $R_{\min}(T_f)$ ’, ‘relation’: ‘=’, ‘quantity’: ‘ $\Theta(\log^2 n)$ ’}

Rationale (proposer’s reasoning):

{‘text’: ‘Tropical geometry provides a natural framework to study polynomial functions, and tensor products have been used to understand the complexity of functions in algebraic geometry. By examining the minimal rank of the tropicalized tensor product matrix, we may uncover new structural properties that could lead to ACC⁰ lower bounds.’, ‘invariant_explanation’: ‘The invariant $R_{\min}(T_f)$ captures the geometric structure of f in the tropical setting and potentially reveals the complexity of evaluating f on a circuit.’}

Taxonomy category: ACC_SIPSER (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): da1f87a5b11b57cd

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all functions f in $\mathcal{P}$ with n variables and for at least 80% of the 30 random seeds, $R_{\min}(T_f) \geq \Omega(\log^2 n)$. The conjecture is falsified if any seed results in $R_{\min}(T_f) < \Omega(\log^2 n)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.8s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = 30
    f = lambda x: sum([x**i for i in range(n)])
    T_f = [[f(2**i)] * n for i in range(n)]

    def tensor_product_rank(matrix):
        m, k = len(matrix), len(matrix[0])
        rank = 0
        for col in zip(*matrix):
            if any(col[i] != 0 for i in range(m)):
                rank += 1
        return rank

    R_min_T_f = tensor_product_rank(T_f)

    metric_name = "minimal_rank"
    metric_value = R_min_T_f
    instances_tested = 1
    conjecture_holds = R_min_T_f >= math.log2(n) ** 2
    counterexample = "" if conjecture_holds else f"R_min(T_f) = {R_min_T_f}, expected  $\geq \log^2(\{n\})$ 

    return {
        "metric_name": metric_name,
        "metric_value": metric_value,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }
```


value of 30 could be an artifact of the specific function or parameter chosen, rather than a general property.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test has only been performed on a single instance ($n \leq 15$), which is insufficient to validate the conjecture for larger values of n . The critic challenged the results, suggesting that the metric value of 30 could be an artifact of the specific function or parameter chosen. | next: Perform additional tests with a variety of functions and parameters to verify the conjecture across different instances.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12474
2	preregistration	ollama_remote	glm4:latest	0	0	5654
3	novelty	ollama_remote	glm4:latest	0	0	4931
4	novelty	ollama_remote	glm4:latest	0	0	5271
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	42625
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7739
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10484
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	20776
9	critic	ollama_remote	glm4:latest	0	0	53288
10	judge	ollama_remote	glm4:latest	0	0	6517

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 169758 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/019b1dc02678.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/019b1dc02678.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/019b1dc02678.tar.gz (if generated)